

Norms

Small Gain Theorem

- ↳ Intuitive: A way of measuring the "size/magnitude" of signals and systems
- ↳ Matrix A can project an arbitrary vector x onto another vector $y \rightsquigarrow y = Ax$
- ↳ Singular Value decomposition (SVD) tells us, which input in x (or direction) gets amplified the most/least

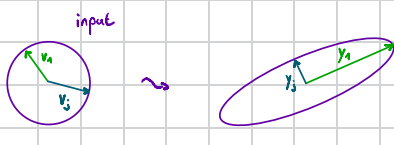
↳ How to do a SVD? Example: $A = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$

Step 1: Eigenvalue AA'

Step 2: Eigenvector AA'

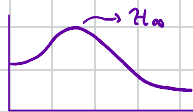
Step 3: Eigenvector AA'

Step 4: Construct $A = U \Sigma V'$

- ↳ Now with the outputs, what does it mean?
 - ↳ We rewrite $A = U \Sigma V' \rightsquigarrow AV = U \Sigma \rightsquigarrow AV_i = \sigma_i u_i$
 - ↳ Now, for each input direction V_i , we have output direction u_i and magnitude σ_i
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$y_1 = \sigma_1 u_1 = \bar{\sigma} u_1$ (Strongest amplification)
 $y_2 = \sigma_2 u_2 = \underline{\sigma} u_2$ (Weakest amplification)
- ↳ The maximum Singular Value $\bar{\sigma}$ represents the maximum amplification in our system (Induced 2-Norm, see later)

- ↳ As mentioned before, we see signals x, y as vectors, so we can set some norms to "measure" them
- ↳ ∞ -Norm $\|u\|_\infty = \sup_{t \in T} \|u(t)\|_\infty = \sup_{t \in T} \max_{i=1, \dots, n} |u_i(t)|$ $\rightsquigarrow$ The absolute maximum a signal can reach at any point in time (Peak magnitude)
- ↳ 2-Norm $\|u\|_2 = \left\{ \begin{array}{l} \sum_{k \in Z} u[k]^T u[k] = \sum_{k \in Z} \|u[k]\|_2^2 \quad (DT) \\ \int_{-\infty}^{\infty} u(t)^T u(t) dt = \int_{-\infty}^{\infty} \|u(t)\|_2^2 dt \quad (CT) \end{array} \right\}$ Square root of total energy over time
- ↳ Example:
 -  Drone forever hovering slightly with small energy magnitude
 - ↳ $\|u\|_2 \rightarrow \infty$ ↳ $\|u\|_\infty \rightarrow \bar{w}$
 -  Drone hit by lightning, a short, but very high magnitude signal input
 - ↳ $\|u\|_2 \rightarrow 0$ ↳ $\|u\|_\infty \rightarrow \infty$

- ↳ Now we can look at systems, remember how SVD gave us the maximum singular value?
- ↳ The $\mathcal{L}_2$ -induced norm, or $\mathcal{H}_\infty$ norm is the maximum singular value over all t !
- ↳ Imagine for SISO system, $\mathcal{H}_\infty$ norm is the peak of the bode plot 
- ↳ We also have a way to calculate the system's energy
 - ↳ The $\mathcal{H}_2$ -Norm measures the energy of the system's impulse response ($\mathcal{H}_2$ correlates with LQG!)
 - ↳ Another interpretation is seeing $\mathcal{H}_2$ as the expected average energy of a white noise

↳ Example: 

Suspension in a car as our model

- ↳ $\mathcal{H}_2 \rightsquigarrow$ Driving on a gravel road, slightly bumpy, but expected 
- ↳ $\mathcal{H}_\infty \rightsquigarrow$ Driving on a bad speed bump, worst case 

↳ Calculation of $\mathcal{H}_2$

System: $A = \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix}$ $B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ $C = \begin{bmatrix} 2 & 1 \end{bmatrix}$

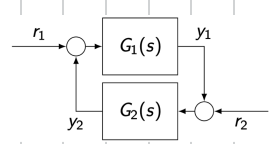
↳ Calculation via Lyapunov equation

- ↳ $AR + RA' = BB'$ or $A'O + OA = C'C$
- ↳ $\|G\|_{\mathcal{H}_2}^2 = \text{Tr}[CRC'] = \text{Tr}[B'OB]$

↳ Calculation of $\mathcal{H}_\infty \rightsquigarrow$ No analytical way!

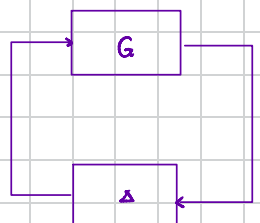
- ↳ We check $\|G\|_{\mathcal{H}_\infty} < \gamma$ for $\gamma > 0$

- ↳ Based on $\mathcal{H}_\infty$, we can derive a sufficient condition for stability for a system like this


$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} (I - G_1 G_2)^{-1} G_1 & (I - G_1 G_2)^{-1} G_1 G_2 \\ (I - G_2 G_1)^{-1} G_2 G_1 & (I - G_2 G_1)^{-1} G_2 \end{bmatrix} \begin{bmatrix} r_1 \\ r_2 \end{bmatrix}$$

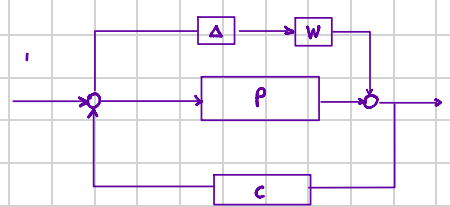
- ↳ We know the interconnection is stable, if
 - ↳ G_1/G_2 are both stable
 - ↳ $\|G_1\|_{\mathcal{H}_\infty} \cdot \|G_2\|_{\mathcal{H}_\infty} < 1$

Applies even to non-linear/time-varying systems. But only a sufficient condition!



Robust Control

- ↳ So let's put the Small gain theorem in action
- ↳ We have our Plant P , with controller C , and a small uncertain block Δ which is applied to a frequency dependent weight W



- ↳ We only know $\|\Delta\|_{\mathcal{H}_\infty} < 1$, how can we ensure that the system stays stable?
- ↳ We simplify the entire system, so $G = -(I + C(s)P(s))^{-1} C(s)W(s)$, just like in example above, we now only need $\|G\|_{\mathcal{H}_\infty} < 1$

Modern Control Design

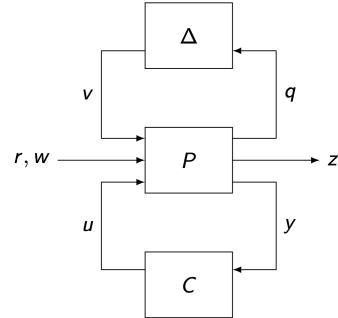
- ↳ With robust control in mind, we can set up a "template" to illustrate all systems

↳ Example: Rocket

Diagramm	"Rocket-Term"
Plant P	
Controller C	
Uncertainty Δ	
r, w	
u	
y	
z	
v, q	



- ↳ Goal: (i) Stable system even with uncertainty
- (ii) Low control effort
- ↳ Design strategies
 - ↳ $\mathcal{H}_2$ /LQG, minimizes total energy output over time
 - ↳ $\mathcal{H}_\infty$, minimizes max energy output



Some exam tasks (Feb. 2025)

Question 17 Mark all correct statements. (2 Points)
Which of the following statements about the norms are true?

☐ A The $\mathcal{H}_\infty$ norm is always greater than or equal to the $\mathcal{H}_2$ norm: $\|G\|_{\mathcal{H}_2} \leq \|G\|_{\mathcal{H}_\infty}$.

☐ B The $\mathcal{H}_\infty$ norm is an induced norm, and for any $G_1, G_2 \in \mathcal{H}_\infty$, $\|G_1 G_2\|_{\mathcal{H}_\infty} \leq \|G_1\|_{\mathcal{H}_\infty} \|G_2\|_{\mathcal{H}_\infty}$.

☐ C The $\mathcal{L}_2$ -induced norm of a system is its $\mathcal{H}_2$ norm, i.e. $\sup_{u \neq 0} \frac{\|y\|_{\mathcal{L}_2}}{\|u\|_{\mathcal{L}_2}} = \sqrt{\frac{1}{2\pi} \int_{-\infty}^{\infty} \|G(j\omega)\|_F^2 d\omega}$, where u is the input signal, y is the output signal, $\|G(j\omega)\|_F^2$ is the Frobenius norm of $G(j\omega)$.

☐ D The $\mathcal{H}_2$ minimization coincides with the LQE and LQR problem statement, because $\mathcal{H}_2$ is an induced norm.

Question 18 *Mark the correct answer. (2 Points)*
Given the transfer function of a system at a certain frequency

$$P = \begin{bmatrix} 2 & 1 \\ -2 & 1 \end{bmatrix}.$$

What is the least amplified input u direction?

- ☐ A
- $u = \begin{bmatrix} 0 & 1 \end{bmatrix}^\top.$
- ☐ B
- $u = \begin{bmatrix} 1 & 1 \end{bmatrix}^\top.$
- ☐ C
- $u = \begin{bmatrix} -1 & 0 \end{bmatrix}^\top.$
- ☐ D
- None of the others.
- ☐ E
- $u = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}^\top.$
- ☐ F
- $u = \begin{bmatrix} -2 & 2 \end{bmatrix}^\top.$
- ☐ G
- $u = \begin{bmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}^\top.$